

Automated End-To-End Data Pipeline For Online Retail Analytic

Almaas Nadia Putri Jaya
School of Computer Science
BINUS University
Jakarta, Indonesia
almaas.jaya@binus.ac.id

Maria Theresia, Sie
School of Computer Science
BINUS University
Jakarta, Indonesia
maria.sie@binus.ac.id

Elizaveta Falisha Susanto
School of Computer Science
BINUS University
Jakarta, Indonesia
elizaveta.susanto@binus.ac.id

Abstract—Retail companies create a vast amount of transactional data everyday. These raw data can't be analyzed directly since it contains a lot of duplicates, missing values or inconsistent format. This project aims to build an end-to-end data engineering pipeline for the Online Retail II dataset. The dataset used covered around 400,000 transactions from 2010 to 2011 through multiple countries. Pentaho Data Integration handles loading and cleaning the data, including the removal of values that doesn't comply to business logic. PDI also removed deduplication through a HashSet method and ensuring data quality at every stage. The cleaned data is rearranged into a Star Schema with a central sales fact table and four supporting dimensions (customer, product, country, and time). The structured data then saved to SQL database storage and is connected to Power BI for visualization. Data is orchestrated through a Pentaho Job that loads all the dimensions before the fact table. It also executed daily through Kitchen.bat and Windows Task Scheduler. The resulting dashboard shows the business total revenue is \$8.75 million with 5 million units sold. The Paper Craft category is also found as the largest share of profit with United Kingdom leading in overall sales volume. Cohort analysis reveals a sharp drop in customer retention after the first month, ranging from 11% to 36%, while EIRE stands out with the highest average revenue per customer. These findings point to the need for early post-purchase retention campaigns and suggest migrating the pipeline to a distributed framework such as Apache Spark to support future real-time processing needs.

Keywords— Retail analytics, ETL Pipeline, Star Schema, Pentaho Data Integration, Business Intelligence

I. INTRODUCTION

A lot of retail companies are experiencing difficulties nowadays in managing vast amounts of information that is collected from everyday transactions. Since there is a lot of analytical advancement these days, the raw information from everyday sales is no longer enough for supporting decision-making because it lacks organization and is often duplicated [1].

Data Engineering offers methods to build data pipelines that transform sales information into a valuable resource for businesses, thereby enhancing the process of making well-reasoned decisions through data. A clear example of when this is essential is when analyzing the "Online Retail II" dataset that contains cross-border retail transactions from 2010 to 2011. This dataset consists of hundreds of thousands of sales

records, detailing customer IDs, prices, countries, and quantities of products, which must be organized through data engineering to resolve formatting issues so the data can be turned into insightful analyses, such as retention studies.

In typical retail environments, raw transactional data such as the records in the Online Retail II dataset is naturally noisy and frequently unsuitable for direct business analysis. Some common challenges include inconsistent metrics like negative prices and also quantities which could distort revenue reporting, and also missing values like empty Customer IDs or even product descriptions [2]. Data gaps where information is still mostly inconsistent and fragmented usually result from the lack of an organized pipeline implemented in modern enterprise data management [2]. With the lack of a specialized data engineering process, decision-making bottlenecks happen due to analysts having to spend about 80% of their operational time manually cleansing data instead of extracting key business insights [3].

This project's main aim was to create a complete system to handle raw transaction data from the Online Retail II dataset; additionally, this system was designed to clean up the raw transaction data to make it properly structured for analysis. This project uses the ETL method, which stands for Extract, Transform, Load; using Pentaho Data Integration. This strategy can help thoroughly clean the data and prevent any contamination of the primary MySQL tables. With ETL, there is no need to set up large temporary tables in MySQL for the raw data.

In the transformation phase, we categorized the data into a star schema by using fact tables and dimension tables because this setup improves the performance of queries during analysis. Through all these methods, the resulting outputs are a collection of relevant data that is integrated directly into Power BI, as well as further analysis in the form of relevant business insights; because the data quality has been validated to meet business needs.

II. LITERATURE REVIEW

A. The Data Engineering Lifecycle

All advanced analytics and machine learning applications are based on the data engineering lifecycle, the framework that transforms raw data into a dependable and also reliable strategic asset [2]. The main stages of this lifecycle include data generation, ingestion, storage, transformation, and lastly serving. In complex retail environments, this lifecycle is very important for minimizing data barriers and creating tailored assets for historical analysis [2]. Ensuring both high data quality and reliability, which also ensures that the information consumed by stakeholders is both valid and consistent for data-driven decision-making, is a key objective throughout these stages [4].

B. ETL (Extract, Transform, Load)

ETL is a basic data integration methodology designed to extract data from raw source systems, apply the necessary transformations, and load the processed data into a centralized Data Warehouse [2]. Raw data gets ingested directly from source files during the extraction phase. In order to ensure that the data is formatted for analysis, the transformation phase is essential for cleaning, validating, and aggregating the data, especially handling missing Customer IDs and standardizing formats [5].

In the process of the load phase, the data is deposited into a target SQL database with a Star Schema structure, facilitating multidimensional OLAP analysis [6]. Since ETL guarantees that transformation logic is strictly, properly executed and validated before the data ever reaches the storage layer which assures high data integrity, it is often preferred in governance-heavy environments over modern ELT approaches [5].

C. Dimensional Modelling

Database design methods known as "dimensional modeling" and "star schema" assist to improve and speed up the process of getting data. A star schema is a way of organizing data; enhancing the speed of queries and making accessing data easier if compared to other designs. Star schemas are often suggested for business intelligence (BI) systems because they can handle quick query results [7]. Their uncomplicated layout also makes them a good option for desktop BI tools like Power BI; which lets users see data visualizations quickly based on their inquiries.

In terms of structure, a star schema consists of two key parts, namely a fact table and a dimension table. The fact table is the central part of this Star Schema design, and it keeps monitoring of numerical data related to business activities, such as the "quantity" feature found in the Online Retail II dataset. An outstanding aspect of the fact table is that its data increase gradually as new transactions happen [8]. On the other hand, dimension tables perform as additional tables that hold descriptive information; giving context to the fact table. These tables can help us to filter and sort data according to

specific traits, like customers, products, countries, and time frames.

D. Orchestration and Automation

According to the research conducted, automation is essential to increase the efficiency and reliability of today's data systems [9]. Automation plays a key role in ETL through minimizing human error and data management. An automated ETL approach reduces the chances of manual errors such as data duplication, errors during transformation and data updates. Also, automated ETL helps to ensure that data management is uniform, thus enabling data warehouse refresh faster and more accurately.

Orchestration is used in the monitoring and unification of ETL pipelines to ensure that they work systematically and automatically [10]. Through the use of Pentaho Data Integration, orchestration takes place through Pentaho jobs defined in transformed processes that are commanded in addition to the ETL kitchen and kitchen jobs to perform in the line using the kitchen. bat, which can be scheduled using Windows Task Scheduler. By using effective orchestration, it ensures successful automated updates of data warehouses without human error.

III. METHODOLOGY

A. Data Source

The information used in this work comes from the Online Retail II Dataset found on Kaggle, which includes data about online shopping sales. The dataset was obtained in CSV form and has around 400,000 records of daily sales from the years 2010 to 2011. It shows actual sales data and has important details like Invoice, StockCode, Description, Quantity, Price, InvoiceDate, Customer ID, and Country. This project uses the data to look into sales activities and create a data warehouse based on a star schema to help with business analysis and data visualization.

B. Ingestion Strategy

The data was taken in using Pentaho Data Integration (PDI) by pulling information from the CSV file into a temporary database (OLTP) before it was processed into the data warehouse. Because the dataset is from the past and is used as a starting load, a complete load ingestion method was used. This method makes sure that all sales data is completely added to the system, keeping the data consistent throughout the ETL process.

Following the transfer to the raw database, the next step of the ETL process was to transfer the data to an OLAP database by preparing a so-called star schema, which consists of dimension tables and fact tables. At this stage, a method of incremental data loading was adopted to optimize the data processing. The method is based on using the etl_metadata table to record information about the last load operation. Consequently, the system only processes new or changed data;

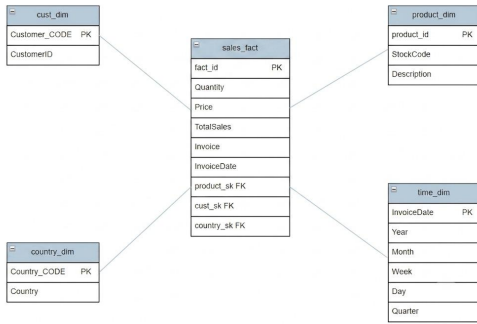
this makes the ETL process faster and more efficient and also saves resources by preventing unnecessary data processing.

C. Data Storage

The way the data is stored is designed in two levels keeping raw data separate from data that's ready to be examined. This separation helps in managing everything more clearly and effectively. The first level is called the "Raw Data Store"; located in the SQL database staging area, where the original data is kept unchanged. This "Raw Data Store" level has several superiorities, such as simplifying data verification, enabling recovery in the event of processing errors, and also keeping a record of changes, so it will be easy to do a traceback if there is an error to look for. Hereafter, the second level is called the "Processed Data Store," which organizes information in a data warehouse to conform to a specific structure created for analysis. When these two levels are combined, Power BI can retrieve data much more quickly and effectively than if retrieved directly from the staging area.

D. Data Modelling

Fig. 1. Star Schema



The modeling technique used to process the data is Star Schema. In this project, Star Schema focuses on transactional data to provide more robust analysis by dividing the data into several main sections; they are sales_fact, cust_dim, country_dim, product_dim, and time_dim.

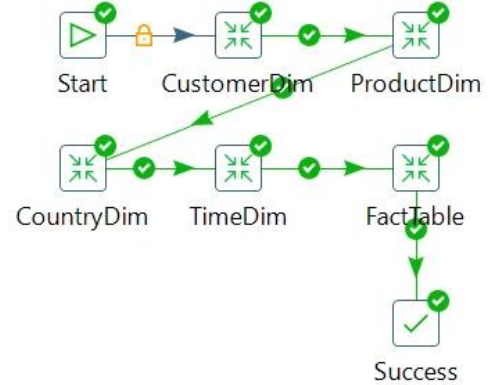
The sales_fact table stores all factual numeric data and has value that can be analyzed using dimension tables, for example, to calculate the number of items sold, the price per unit, and the total revenue. In addition to storing numeric data, sales_fact also has a foreign key to connect to all dimension tables.

Meanwhile, the dimension table is further divided into four tables to simplify sorting, sharing, and summarizing basic data tailored to business needs. These four tables are cust_dim, country_dim, product_dim, and time_dim. First, the cust_dim table contains customer information to help understand purchasing patterns, while the product_dim table contains the detailed information for each product; such as stock quantities and product descriptions. Second, for location analysis, country_dim shows where each transaction occurs. And finally, the time_dim table is used to separate dates (time granularity) into single days, months, and years; which allows us to focus on and study sales trends that happen during certain seasons or time periods.

E. Automation and Orchestration Design

To minimize human error and fulfill almost every task without the need for manual handling, the ETL workflow relies on a completely automated orchestration design [4]. Pentaho Data Integration links individual transformation files in (.ktr) and manages the execution sequence using a master Pentaho Job (.kjb) like in the figure below.

Fig. 2. Pentaho Job



The use of Pentaho Job like in the figure above basically protects referential integrity by making sure that the dimension tables such Customer, Product, Country, and Time have been completely filled before processing the Sales Fact table [6]. For the pipeline to be able to achieve automation through a Command Line Interface (CLI), Pentaho's own batch script (Kitchen.bat) that is integrated into Windows Task Scheduler is used to start up the pipeline once daily [4]. This automated orchestration guarantees that the Data Warehouse and downstream Power BI dashboards can regularly get the most recent data without needing manual user intervention [4].

IV. IMPLEMENTATIONS & RESULTS

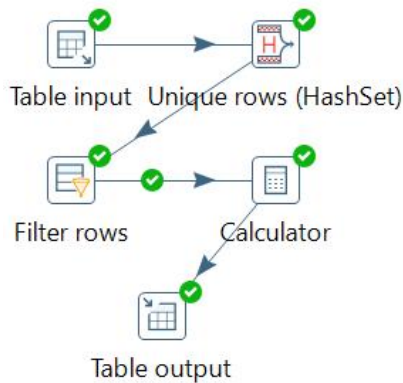
A. Data Transformation & Quality

The cleansing process consists of standardizing string formats and removing unwanted special characters from the Description field like commas and horizontal tabs in order to avoid downstream processing errors. Pentaho's data transformation step applies strict business rules and data quality checks across many different transformation files in (.ktr) before data reaches the Data Warehouse. This integrated method enables the incremental correction of data anomalies before implementing dimensional modeling [11]. The following core .ktr components form the transformation lifecycle procedures:

1) Initial Cleansing and Fact Preparation (Transformation 1)

The first transformation extracts raw records straight from the OLTP staging table so it serves as the main data integrity checkpoint.

Fig. 3. The Staging Table

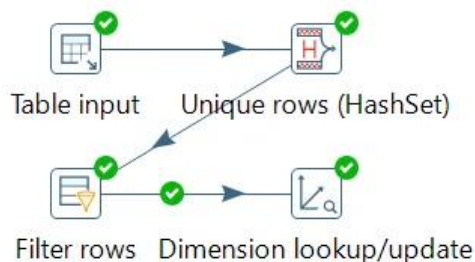


To minimize dataset redundancy, the duplicate transactions are actively identified and removed using a Unique Rows (HashSet) step based on exact row matches. After doing so, unstructured or inaccurate operational data is removed using strict business logic in a Filter Rows step. Specifically transactions with a Price of zero or less or a Quantity of zero or less are removed entirely. These anomalies usually point to unorganized return logs instead of actual sales. To reduce the computational load (DAX operations) on the downstream Power BI dashboard, the Calculator step multiplies Quantity and Price to derive a TotalSales feature before the data gets committed to the cleaned staging table.

2) Dimension Processing & Standardization (Transformations 2 & 3)

Using tailored transformations, the pipeline creates unique, better dimension tables such as Product and Customer.

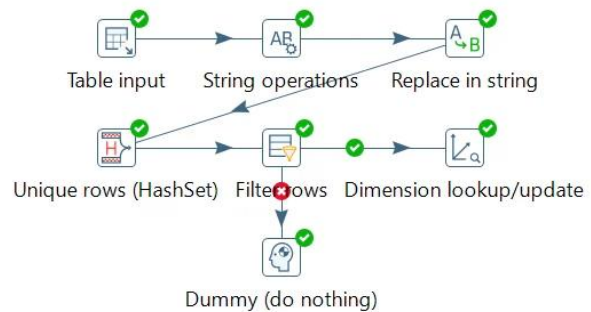
Fig. 4. Customer Dimension Transformation



Since string data in retail operations is typically inconsistent, the Customer Dimension transformation enforces strict formatting constraints and data completeness checks to ensure that only accurate data moves to the Data Warehouse without affecting the overall integration flow. The Filter Rows step is used to remove any record with a null Customer ID so incomplete keys do not enter the dimension as they might make the customer-based analysis invalid. The last step that is used is Dimension Lookup/Update which applies SCD Type 1 logic to write the data into the dimension database and create

surrogate keys. This means that if a customer's characteristics change, the current record simply gets replaced instead of it being kept as a history of previous values. The final result is a dimension table that remains clean and preserves the uniqueness of its primary key before the fact table makes reference to it.

Fig. 5. Product Dimension Transformation

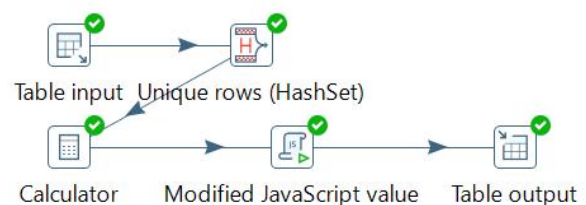


The String Operations step is used Type (Both) to the Description and StockCode fields in addition to converting the StockCode to uppercase for complete uniformity because string data in retail is often inconsistent. The Replace in String step is used next to forcefully remove the horizontal tabs and commas from the Description field to normalize characters that may impact formatting downstream. Completeness of data is strictly enforced, so records lacking a StockCode or Description are routed out of the main pipeline and into a Dummy (do nothing) step that uses a secondary Filter Rows as a quality gate, isolating incorrect data without causing the pipeline to crash. And since this dimension used SCD type 2, a new surrogate key (product_code) is created to track the changes.

3) Time Dimension Extraction (Transformation 5)

To be able to make periodic trend analysis, the Time Dimension table is created through a tailored transformation.

Fig. 6. Time Dimension Transformation

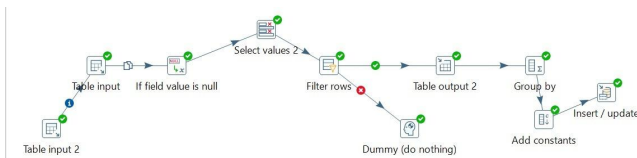


To get unique InvoiceDate values, the Unique Rows is used, and after that the data goes through the Calculator step to automatically extract discrete temporal elements such as Year, Month, Week, and Day. To classify the fiscal quarters, custom scripting is run using the Modified JavaScript Value step. Since the calendar data is structurally static, this transformation uses the normal Table Output step configured with the Truncate method rather than complex lookups. This guarantees that the dimension stays properly in sync with the most recent and current daily transactions.

4) Metadata and Incremental Load Strategy

An incremental load method is directly modeled to guarantee pipeline efficiency for ongoing operations.

Fig. 7. Fact Transformation Using Incremental Strategy

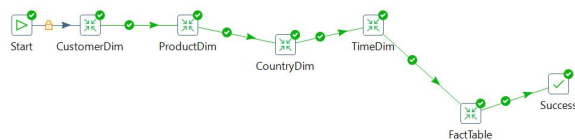


By mapping the data with the Group By the MAX function, the pipeline identifies the date of the most recent transaction that was loaded into the system. The Add Constants step provides a primary key for the metadata control table and the Insert/Update step comes after. This method greatly reduces server load and processing time by ensuring that only newly generated or changed transaction records will be dynamically queired and processed in subsequent ETL executions by keeping track of the Last Load Date.

B. Automation and Orchestration

1) Job Orchestration using Pentaho Jobs

Fig. 8. Pentaho Job Orchestration



Regarding the ETL pipeline workflow, all transformation files .ktr) are collected into a single Pentaho project file .kjb) to facilitate process coordination. This Pentaho project manages the execution sequence in an organized and integrated manner. Specifically, all dimension tables – including ProductDim, CustomerDim, CountryDim and DateDim were loaded before the fact table SalesFact was populated. This approach promotes data consistency and ensures that all necessary foreign keys are present in the fact table before it is populated.

2) CLI Execution via Batch Scripting

The ETL process is run directly from the command line using Kitchen.bat. For this purpose, a batch file (.bat) has been prepared, which fulfills the task of opening and executing the .kjb file created with Pentaho Kitchen without the need to click on Spoon. Before starting these processes, the script also prepares the Java environment, which is necessary to complete the process successfully. The script used to run this automation is written below:

```
@echo off
```

```
title Eksekusi Job Maria Thereisa
```

```
echo [1] Setting Java JDK 25...
```

```
SET "JAVA_HOME=C:\Program Files\Java\jdk-25.0.2"
```

```
SET "PATH=%JAVA_HOME%\bin;%PATH%"
```

```
echo [2] Masuk ke Folder Launcher...
```

```
cd /d "D:\Software\data-integration\launcher"
```

```
echo [3] Menjalankan Job...
```

```
java -jar launcher.jar -main org.pentaho.di.kitchen.Kitchen -  
file "D:\SEMESTER 4\Job 1.kjb" -level Basic
```

```
echo.
```

```
echo [4] Selesai!
```

```
pause
```

The script launches Pentaho Kitchen directly from the command line and runs the job file that was created. This script has an advantage because the job can run without needing to open the Spoon application. The ETL process can run by itself with this script. It does not need someone to do anything for it to work. This automation makes the ETL process a lot easier to manage.

3) Scheduling with Windows Task Scheduler

The batch file is scheduled to run using the Windows Task Scheduler to automate the entire process from start to finish without human intervention. The computer executes the script automatically every day at 19:35 using this setup. This made the ETL process work completely by itself since the process is already automated and it makes the data update process more organized and efficient, while reducing the risk of human error associated with manual pipeline execution. This automation also ensures that the data warehouse and the Power BI dashboard will always have the up to date data (the latest data).

Fig. 9. Job Automation Implementation in CLI

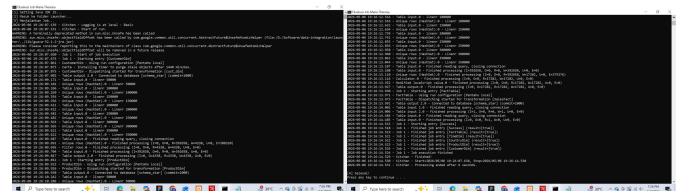
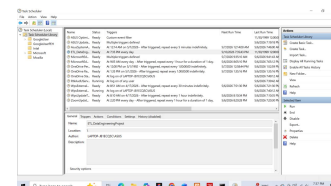


Fig. 10. Automation Implementation with Windows Task Scheduler



The automation and orchestration implementation works well as shown in the attachment. The batch file starts the Pentaho

job using Kitchen without any issues then Windows Task Scheduler runs the pipeline at the time it is supposed to. We know this is working because the Pentaho Kitchen logs say "Finished" and the Windows Task Scheduler logs that show the saved task and the triggers. This means the ETL pipeline succeed in running by itself automatically without anyone having to do anything.

C. Data Consumption

The last step in the data engineering workflow is data consumption. It is including presenting data from the data warehouse with an engaging Power BI dashboard. In this step, transactional data needs some changes into meaningful insights for making decisions. The visualization breaks down into three key parts and each of them showing different sales data.

Fig. 11. Sales Performance Dashboard

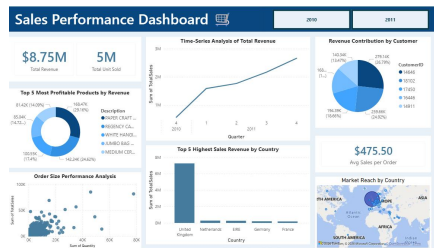
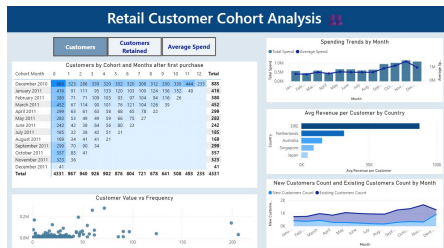


Figure 11 illustrates the dashboard for monitoring overall sales data;featuring important indicators like total sales of \$8.75 million, total units sold of 5 million, and average income per order of \$475.50. This visualization offers a number of strategic understandings about the company's sales. The sales pattern shows a constant rise from the end of 2010 to a high point in the last quarter of 2011. Next, the "Paper Craft" product emerges as the leading product, contributing nearly 28% of the income from the top five items. As can be seen in the "Order Size Performance Analysis" graph, the majority of sales come from smaller retailers, although larger orders do occur from time to time. The evidence that the majority of sales come from smaller retailers is indicated by the large number of dots clustered in the bottom left corner; this indicates that customers are significantly more likely to purchase smaller orders than larger orders. Finally, the "Market Reach by Country" graph highlights the company's global reach; with particularly strong sales in Europe, especially the United Kingdom.

Fig. 12. Retail Customer Cohort Analysis



The "Retail Customer Cohort Analysis" dashboard in Figure 12 shows customer loyalty and spending habits using three key metrics; they are Customers by Cohort, Spending Trends, and Average Revenue per Customer.

The "Customers by Cohort and Months after First Purchase" graph shows significant differences in customer loyalty between months, particularly a decline in the first month; all cohorts experienced a significant decline in the number of active customers after their first purchase. This impacted retention rates, which ranged between 11% and 36%. This significant decline in customer loyalty suggests that customer loyalty programs or special offers, such as welcome programs or soft opening programs, motivate customers to shop, and therefore need to be reintroduced to attract return customers. Meanwhile, the December 2010 cohort shows a significant number of loyal customers, retaining over 300 customers in the tenth month and increasing to 444 customers in the eleventh month. From this pattern in the December 2010 cohort, it can be concluded that end-of-year promotions during the holidays successfully retained existing customers.

Over and above that, based on the "Spending Trends by Month" graph, it can be seen that the total revenue remained relatively stable from January to September, then it's finally surging from October to December. Among the months that experienced the surge, there was November where total spending reached its highest figure, which is \$1 million. However, it's important to note that despite the sharp increase in revenue at the end of the year, the average transaction value remained the same. This means that the increase in revenue in that quarter (Q4 peak) was mainly due to more transactions and a larger customer base, not due to price increases or an increase in average order value.

Furthermore, through the "Avg Revenue per Customer by Country" chart, it can be seen that EIRE holds the strongest position in the market because the average revenue generated per customer is close to \$100,000, making EIRE a strong contender for future loyalty initiatives. EIRE's position as a strong market also places other countries such as the Netherlands, Australia, Singapore, and Japan, where the average revenue is just under \$50,000, far behind EIRE.

Finally, the "New Customer Count and Existing Customer Count by Month" graph shows that it was the number of existing customers that increased, not new ones. The number of existing customers increased further at the end of the year, indicating that a successful rewards program is essential for the company's continued success.

Fig. 13. Retail Product Sales Analysis

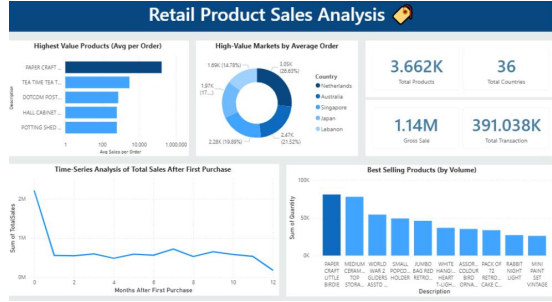


Figure 13 provides a summary of worldwide production and market distribution, utilizing several important macroeconomic metrics; total products (3,662,000), total countries (36), overall earnings (\$1.14 million), and the number of transactions (391,038,000). This figure outlines key insights related to overall sales characteristics. Initially, the Netherlands leads the global market, boasting the highest average selling price and holding a 26.63% share of the entire market. Additionally, the "Paper Craft Little Birdie" product is noteworthy, as it leads in two areas; average price per item and total sales volume. The data also shows that the selling price dropped notably after the first month, then leveled off at \$0.5 million by the eleventh month. Lastly, the chart in the bottom right displays the top-selling items, with the top ten making up the overall total.

V. EVALUATION AND REFLECTION

In this project, the evaluation of the software looked at three main points; how well it works, how easily it can grow, and how simply its parts can fit together. Pentaho Data Integration (PDI) was picked as the main ETL tool, mainly because it has an easy-to-use, visual, and drag-and-drop setup that does not slow down the database. This setup greatly sped up the creation of complicated data changes, such as cleaning data, checking data types, and dealing with missing information in hundreds of thousands of rows from the Online Retail II database.

In addition to Pentaho Data Integration, MySQL is also used for various operational needs, such as managing databases, organizing storage into two layers, and using indexing features. The choice of MySQL for database management is based on the fact that MySQL is highly capable of handling structured data.

As mentioned earlier, MySQL is used to organize the storage space into two main layers; the first layer functions as a temporary storage area (staging area) that maintains the security of initial data, while the second layer acts as a data warehouse to store data that has been through the process of being modified. Not only to organize the storage, but also the indexing feature in MySQL is put into practice to speed up query processing, thus providing faster and more efficient access to the fact_sales fact table and related dimension tables.

VI. CONCLUSION

A. Conclusion

All in all, in the end this project has successfully developed an end-to-end ETL pipeline using Pentaho Data Integration to process raw data from the Online Retail II dataset. This data engineering process was designed to immediately cleanse the raw data and insert it into a MySQL star schema; consisting of fact and dimension tables, so it eliminates the need for large temporary tables.

After processing the raw data using Pentaho Data Integration, the dashboards and macro sales data analysis at the Data Consumption phase prove that the cleaned dataset in Pentaho can be dynamically connected to Power BI to produce interactive visualizations and to analyze sales trends, such as those that occurred in the fourth quarter of 2011; when revenue reached \$8.75 million with sales volume reaching 5 million units; then the "Paper Craft" product category became the largest revenue contributor, and a very strong market concentration was in the Europe region, especially the UK.

Even though these macro sales trends seem auspicious, analysis of loyalty cohorts and customer behavior reports significant vulnerabilities in the current business model. Where nearly all cohorts experienced a sharp decline in retention, ranging from 11% to 36%, within the first month after their initial transaction. Meanwhile, an engrossing trend was found only in the December 2010 customer cohort, which maintained consistent transaction activity through the eleventh month, which suggest that year-end promotions were far more effective in increasing transaction volume and expanding the active customer base than in increasing average order value.

Based on "Retail Customer Cohort Analysis" dashboard earlier, EIRE has the highest average revenue per customer, which is nearly \$100,000, while the Netherlands has the highest average purchase price, which is 26.63%. Meanwhile, "Paper Craft Little Birdie" is has successfully ranked at the top, both in terms of average price per unit and total quantity sold in the market.

B. Recommendations

Given the development's technological limits and the business insights gleaned from the data, several suggestions for future work and long-term strategic direction can be made. From a company and customer loyalty standpoint, the dramatic drop in customer activity after the first month necessitates the implementation of tailored loyalty programs and automatic post-purchase marketing. Given the high average return per customer data from EIRE, this market should be prioritized as a major testing ground for future loyalty strategies. Furthermore, while the volume rise in the fourth quarter was effective, promotional methods similar to those used in the campaigns that ended in 2010 should be implemented to sustain more steady transaction volumes in the first quarters of 2011.

While Pentaho Data Integration has successfully processed the current data set, future development should consider migrating the ETL pipeline to a distributed processing framework such as Apache Spark, especially to support larger volumes of real-time or streaming data transactions.

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